ADJUSTABLE-SPEED PERMANENT MAGNET SYNCHRONOUS MOTOR DESIGN

File: Setup1.res

GENERAL DATA

Rated Output Power (kW):	0.055
Rated Voltage (V):	48
Number of Poles:	8
Frequency (Hz):	100
Frictional Loss (W):	12
Windage Loss (W):	0
Rotor Position:	Inner
Type of Circuit:	L3
Type of Source:	Sine
Domain:	Time
Operating Temperature (C):	50

STATOR DATA

Number of Stator Slots:	15
Outer Diameter of Stator (mm):	100
Inner Diameter of Stator (mm):	62
Type of Stator Slot:	1
Stator Slot	
hs0 (mm):	0.5
hs2 (mm):	3.35354
bs0 (mm):	3
bs1 (mm):	8.45603
bs2 (mm):	9.88167
Top Tooth Width (mm):	6.45584
Bottom Tooth Width (mm):	6.45584
Skew Width (Number of Slots):	0
Length of Stator Core (mm):	100
Stacking Factor of Stator Core:	0.95
Type of Steel:	M19-24G
Designed Wedge Thickness (mm):	1.64807
Slot Insulation Thickness (mm):	0.3
Layer Insulation Thickness (mm):	0.3
End Length Adjustment (mm):	0
Number of Parallel Branches:	1
Number of Conductors per Slot:	2
Number of Layers:	2
Winding Type:	Editor
Average Coil Pitch:	1
Number of Wires per Conductor:	19
Wire Diameter (mm):	1.15
Wire Wrap Thickness (mm):	0
Slot Area (mm^2):	98.1202
Net Slot Area (mm^2):	72.5704
Limited Slot Fill Factor (%):	75
Stator Slot Fill Factor (%):	69.25
Coil Half-Turn Length (mm):	112.024
Wire Resistivity (ohm.mm^2/m):	0.0217

ROTOR DATA

Minimum Air Gap (mm):	1
Inner Diameter (mm):	38
Length of Rotor (mm):	100
Stacking Factor of Iron Core:	0.95
Type of Steel:	M19-24G
Polar Arc Radius (mm):	30
Mechanical Pole Embrace:	0.7
Electrical Pole Embrace:	0.694097
Max. Thickness of Magnet (mm):	4
Width of Magnet (mm):	15.3938
Type of Magnet:	XG196/96
Type of Rotor:	1
Magnetic Shaft:	No

PERMANENT MAGNET DATA

Residual Flux Density (Tesla):	0.96
Coercive Force (kA/m):	690
Maximum Energy Density (kJ/m^3):	183
Relative Recoil Permeability:	1
Demagnetized Flux Density (Tesla):	0.166361
Recoil Residual Flux Density (Tesla):	0.913527
Recoil Coercive Force (kA/m):	726.983

MATERIAL CONSUMPTION

Armature Wire Density (kg/m^3):	8900
Permanent Magnet Density (kg/m^3):	7800
Armature Core Steel Density (kg/m^3):	7650
Rotor Core Steel Density (kg/m^3):	7650
Armature Copper Weight (kg):	0.590282
Permanent Magnet Weight (kg):	0.384229
Armature Core Steel Weight (kg):	2.44414
Rotor Core Steel Weight (kg):	0.719193
Total Net Weight (kg):	4.13784
Armature Core Steel Consumption (kg):	5.98346
Rotor Core Steel Consumption (kg):	1.72663

STEADY STATE PARAMETERS

Stator Winding Factor:	0.71095
D-Axis Reactive Inductance Lad (H):	9.73857e-07
Q-Axis Reactive Inductance Laq (H):	9.73857e-07
D-Axis Inductance L1+Lad (H):	2.97608e-06
Q-Axis Inductance L1+Laq (H):	2.97608e-06
Armature Leakage Inductance L1 (H):	2.00222e-06
Slot Leakage Inductance Ls1 (H):	1.70952e-06
End Leakage Inductance Le1 (H):	4.07919e-08
Harmonic Leakage Inductance Ld1 (H):	2.51908e-07
Zero-Sequence Inductance L0 (H):	1.59436e-06
Armature Phase Resistance R1 (ohm):	0.00113243
Armature Phase Resistance at 20C (ohm):	0.00101323

NO-LOAD MAGNETIC DATA

Stator-Teeth Flux Density (Tesla):	1.39399
Stator-Yoke Flux Density (Tesla):	0.738681
Rotor-Yoke Flux Density (Tesla):	0.833609
Air-Gap Flux Density (Tesla):	0.643194
Magnet Flux Density (Tesla):	0.72635
Stator-Teeth By-Pass Factor:	0.00155227
Stator-Yoke By-Pass Factor:	9.02262e-06
Rotor-Yoke By-Pass Factor:	9.74415e-06
Stator-Teeth Ampere Turns (A.T):	9.95903
Stator-Yoke Ampere Turns (A.T):	1.28368
Rotor-Yoke Ampere Turns (A.T):	0.735595
Air-Gap Ampere Turns (A.T):	583.894
Magnet Ampere Turns (A.T):	-595.821
Leakage-Flux Factor:	1
Correction Factor for Magnetic Circuit Length of Stator Yoke:	0.707551
Correction Factor for Magnetic Circuit Length of Rotor Yoke:	0.679369
No-Load Line Current (A):	43546.9
No-Load Input Power (W):	2.14748e+06
Cogging Torque (N.m):	0.0117003

FULL-LOAD DATA

Maximum Line Induced Voltage (V):	2.55048
Root-Mean-Square Line Current (A):	43546.7
Root-Mean-Square Phase Current (A):	25141.6
Armature Thermal Load (A^2/mm^3):	4.93318e+06
Specific Electric Loading (A/mm):	3872.34
Armature Current Density (A/mm^2):	1273.96
Frictional and Windage Loss (W):	12
Iron-Core Loss (W):	7.41666
Armature Copper Loss (W):	2.14744e+06
Total Loss (W):	2.14746e+06
Output Power (W):	56.0131
Input Power (W):	2.14752e+06
Efficiency (%):	0.00260828
Synchronous Speed (rpm):	1500
Rated Torque (N.m):	0.356591
Torque Angle (degree):	-36.8006
Maximum Output Power (W):	49634.6

WINDING ARRANGEMENT

Edited Winding Layout:

Coil	Phase	Turns	SlotIn	SlotOut

Coil_1	A	1	1T	2B
Coil_2	B	1	2T	3B
Coil_3	-A	1	3T	4B
Coil_4	-B	1	4T	5B
Coil_5	A	1	5T	6B
Coil_6	B	1	6T	7B
Coil_7	C	1	7T	8B
Coil_8	-B	1	8T	9B
Coil_9	-C	1	9T	10B
Coil_10	B	1	10T	11B
Coil_11	C	1	11T	12B
Coil_12	A	1	12T	13B
Coil_13	-C	1	13T	14B
Coil_14	-A	1	14T	15B
Coil_15	C	1	15T	1B

Periodic Multipliers: 1

Angle per slot (elec. degrees): 96
Phase-A axis (elec. degrees): 48
First slot center (elec. degrees): 0

TRANSIENT FEA INPUT DATA

For Armature Winding:
- Number of Turns: 5
- Parallel Branches: 1
- Terminal Resistance (ohm): 0.00113243
- End Leakage Inductance (H): 4.07919e-08

2D Equivalent Value:
- Equivalent Model Depth (mm): 100
- Equivalent Stator Stacking Factor: 0.95
- Equivalent Rotor Stacking Factor: 0.95
- Equivalent Br (Tesla): 0.913527
- Equivalent Hc (kA/m): 726.983